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Attestation of Primary Authorship* and AI Disclosure Statement

We, **Ralph Paolo Dulce and Yahyah Odin**, the proponents of thesis entitled **"A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference"**, hereby declare that:

- We are the sole authors of this thesis and its technical output.
- We have fully acknowledged and referenced the ideas, techniques and work of others, whether published or unpublished, in our thesis.
- We have prepared the thesis specifically for the degree of **Bachelor of Science in Computer Science** while under supervision at Ateneo de Davao University.
- Our thesis document does not contain work extracted from a thesis, dissertation or research paper previously presented for another degree or diploma at this or any other university.
- Our technical output was not subcontracted to a third party.

AI Disclosure Statement

- We hereby declare that any use of artificial intelligence (AI) tools in the preparation of this thesis and its technical output:

AI Tool Used	Purpose and Description of Use	Sections/Components Where AI Was Used
Google Antigravity / Gemini Models	Assisted as an agentic pair-programmer for Python engineering: scaffolding modular codebase architecture (`src/`), automated execution harnesses, data pipelines for dual-corpus JSONL compilation (27,357 enactments, 178,494 provisions), batch OCR wrappers, and engineering the offline interactive web prototype interface (`demo/`).	Codebase Pipelines (`src/`, `scripts/`) and Interactive Web Prototype (`demo/`)
Google Antigravity / Gemini Models	Assisted in LaTeX typesetting, multi-pass compilation orchestration (`pdflatex`/`bibtex` in `scripts/build_paper.py`), resolution of build diagnostics, formatting dedicated full-page landscape plates (`pdflscape`) for empirical tables, managing front-matter listings, and synchronizing the automated single-source numbers registry.	Thesis Manuscript (`main.tex`, Chapters 1–5), Appendices A–P, and Numbers Registry
Google Antigravity / Matplotlib / Seaborn	Assisted in scientific data visualization and publication graphics: scripting 300-DPI figure generation for longitudinal word count evolution, 2D SVD latent semantic topic space with covariance ellipses, 94.75% 512-token compliance curves, Stage 1 Recall@k charts, training loss dynamics, and token attention attribution heatmaps.	Publication Figures across Chapters 3 & 4, Appendices, and Flowcharts (`figs/`)
Gemini Flash 3.8 (Google)	Assisted in syntactic clause drafting and formal legislative enacting phrase templating of candidate local ordinance provisions under strictly human-specified statutory premises, legal mechanisms, and conflict types (Section 3.5.3), conforming to Davao City Sangguniang Panlungsod and DILG drafting manuals without fabricating legal facts.	Methodology (Section 3.5.3) and Ground Truth Benchmark Formulation (`data/`)

**Adapted from Faculty of Graduate Research, Statement of Authorship and Thesis Length for PhD Students, Victoria University of Wellington, 2009*

AI Tool Used	Purpose and Description of Use	Sections/Components Where AI Was Used
Claude 3.5 Sonnet, ChatGPT, & DeepSeek	Assisted as adversarial simulated peer reviewers and mock defense interrogators: stress-testing scientific claims, identifying narrative overclaiming, and conducting forensic checks on mathematical and statistical consistency (McNemar $2\times contingencytableson = 53$, Friedman ANOVA ranks, paired Wilcoxon tests, Wilson score CIs, and Bayesian low-prevalence screening dynamics).	External Peer Review Simulation and Manuscript Stress-Testing (Chapters 1–5, Abstract, Conclusions)
Claude 3.5 Sonnet / Gemini Models	Assisted in editorial prose refinement, sentence flow transitions, eliminating repetitive phrasing or conversational filler, and ensuring adherence to academic continuous paragraph formatting in strict accordance with departmental writing guidelines.	Manuscript Prose Polish across Chapters 1–5, Abstract, and Appendices
Google Antigravity / Gemini Models	Assisted in authoring defense presentation slides strictly following the department 7×7 rule, structuring word-for-word spoken presenter scripts with foundational visual walkthroughs, and developing the interactive HTML presentation slide viewer (`docs/presentation/slides_viewer.html`).	Oral Presentation Slide Decks and Interactive Presentation Viewer (`docs/presentation/`)
Evaluated ML Models (Research Subjects)	Deep neural cross-encoders (`DeBERTa-v3`, `ModernBERT`, `Legal-BERT`, `all-MiniLM-L6-v2`) and vision-language models (`Qwen2.5-VL-3B-Instruct`) were evaluated strictly as offline empirical machine learning models and experimental research subjects under deterministic evaluation on local LGU hardware, not as generative text authors.	Methodology and Experimental Results (Chapters 3 & 4, Appendices A–P)

We further declare that:

- We have accurately disclosed the use of AI tools in the preparation of this thesis and its technical output.
- We take full responsibility for the accuracy, originality, integrity, and academic quality of the submitted work, including any content generated or assisted by AI tools.
- We have reviewed and verified all AI-generated or AI-assisted content incorporated into the thesis and its technical output and have ensured that appropriate acknowledgements and citations are provided, where necessary.

By signing below, we affirm that the statements above are true and accurate.

Signed: _____ **Date:** _____

Ralph Paolo Dulce

Signed: _____ **Date:** _____

Yahyah Odin

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